Entertainment Database — Data Dictionary

Agents

Purpose: Stores all agent information, used to evaluate agent performance, territory coverage, and revenue contribution.

Column Name	Data Type	Description
AgentID	Primary Key (PK)	Unique identifier for each agent
AgtFirstName	Text	Agent’s first name
AgtLastName	Text	Agent’s last name
AgtStreetAddress	Text	Street address
AgtCity	Text	City
AgtState	Text	State abbreviation (e.g., CA, NY)
AgtZipCode	Text	Postal code
AgtPhoneNumber	Text	Agent’s phone number
DateHired	Date	Date when the agent was hired
Salary	Numeric	Agent’s base salary
CommissionRate	Real	Commission percentage or rate

Entertainers

Purpose: Stores all entertainer information, used to analyze entertainer availability, performance, popularity, and style specialization.

Column Name	Data Type	Description
EntertainerID	Primary Key (PK)	Unique identifier for each entertainer or group
EntStageName	Text	Stage name (public performance name)
EntSSN	Text	Social Security Number (internal use)

EntStreetAddress	Text	Street address
EntCity	Text	City
EntState	Text	State abbreviation
EntZipCode	Text	Postal code
EntPhoneNumber	Text	Phone number
EntWebPage	Text	Website URL
EntEmailAddress	Text	Email address
DateEntered	Date	Date when the entertainer was added to the system

Engagements

Purpose: The core revenue table. Stores every booking, linking customers, agents, and entertainers. Crucial for revenue trend, performance, and customer behavior analysis.

Column Name	Data Type	Description
EngagementNumber	Primary Key (PK)	Unique engagement or contract number
StartDate	Date	Engagement start date
EndDate	Date	Engagement end date
StartTime	Time	Start time of performance
StopTime	Time	End time of performance
ContractPrice	Numeric	Contract amount agreed upon
AgentID	Foreign Key (FK)	References Agents.AgentID
EntertainerID	Foreign Key (FK)	References Entertainers.EntertainerID
CustomerID	Foreign Key (FK)	References Customers.CustomerID

Customers

Purpose: Stores customer demographic and contact information. Useful for segmentation, retention, and market analysis.

Column Name	Data Type	Description
CustomerID	Primary Key (PK)	Unique identifier for each customer
CustFirstName	Text	Customer's first name
CustLastName	Text	Customer's last name
CustStreetAddress	Text	Street address
CustCity	Text	City
CustState	Text	State abbreviation
CustZipCode	Text	Postal code
CustPhoneNumber	Text	Customer's phone number

Musical_Styles

Purpose: Defines all music genres.

Column Name	Data Type	Description
StyleID	Primary Key (PK)	Unique identifier for each music style
StyleName	Text	Name of the musical style (e.g., Jazz, Pop, Rock)

Entertainer_Styles

Purpose: Many-to-many relationship table linking entertainers and their styles.

Column Name	Data Type	Description
EntertainerID	Foreign Key (FK)	References Entertainers.EntertainerID
StyleID	Foreign Key (FK)	References Musical_Styles.StyleID
StyleStrength	Smallint	Rating or strength of entertainer's proficiency in that style (1 = primary style, 2 = secondary, and 3 = tertiary)

Entertainer_Members

Purpose: Links entertainers to their individual members.

Column Name	Data Type	Description
EntertainerID	Foreign Key (FK)	References Entertainers.EntertainerID
MemberID	Foreign Key (FK)	References Members.MemberID
Status	Smallint	Membership status (1 = active, 2 = former or inactive)

Members

Purpose: Stores detailed information on entertainer members.

Column Name	Data Type	Description
MemberID	Primary Key (PK)	Unique identifier for each member
MbrFirstName	Text	Member’s first name
MbrLastName	Text	Member’s last name
MbrPhoneNumber	Text	Phone number
Gender	Text	Gender

Musical_Preferences

Purpose: Stores each customer’s preferred musical styles. Supports segmentation and recommendation.

Column Name	Data Type	Description
CustomerID	Foreign Key (FK)	References Customers.CustomerID
StyleID	Foreign Key (FK)	References Musical_Styles.StyleID
PreferenceSeq	Smallint	Ranking order of customer’s preference (1 = most preferred)

ztblDays

Purpose: These are date dimension tables used for time-series analysis.

Column Name	Data Type	Description
DateField	Date	Each calendar date; used for time-based analysis and reporting

ztblMonths

Purpose: These are date dimension tables used for time-series analysis.

Column Name	Data Type	Description
MonthYear	Text	Combined month-year label (e.g., “Jan 2017”)
YearNumber	Integer	Year (e.g., 2017)
MonthNumber	Integer	Month number (1–12)
MonthStart	Date	First date of the month
MonthEnd	Date	Last date of the month
January–December	Boolean	Optional flags for report logic by month

ztblWeeks

Purpose: These are date dimension tables used for time-series analysis.

Column Name	Data Type	Description
WeekStart	Date	Start date of the week
WeekEnd	Date	End date of the week

ztblSkipLabels

Purpose: Stores the number of label positions to skip before starting a new print sequence.

Column Name	Data Type	Description
LabelCount	Integer	Number of label positions to skip before printing

Entertainment Database – Relationships Overview

Table A	Relationship	Table B	Description
Agents	1–∞	Engagements	Each agent can handle multiple engagements
Entertainers	1–∞	Engagements	Each entertainer can participate in multiple engagements
Customers	1–∞	Engagements	Each customer can book multiple engagements
Entertainers	1–∞	Entertainer_Styles	Each entertainer can have multiple musical styles
Musical_Styles	1–∞	Entertainer_Styles	Each style can be assigned to multiple entertainers
Customers	1–∞	Musical_Preferences	Each customer can have multiple musical preferences
Musical_Styles	1–∞	Musical_Preferences	Each style can appear in many customers' preference lists

Entertainers	1–∞	Entertainer_Members	Each entertainer group can have multiple members
Members	1–∞	Entertainer_Members	Each member can belong to multiple entertainer groups

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Unclear or Ambiguous Fields Identified (with Assumptions for Analysis)

During the data familiarization process, several fields across the dataset were identified as unclear or ambiguous. These fields require clarification from TuneWorks' data engineering or operational teams to ensure accurate interpretation. For the purpose of continuing the analysis in later phases, reasonable working assumptions were made, as detailed below.

1. entertainer_styles.stylestrength

- Ambiguity:
  The column stores categorical values such as 1, 2, and 3, but the dataset documentation does not specify whether 1 represents the strongest association or the weakest.
- Working assumption for analysis:
  We assume that smaller values represent stronger style alignment, with 1 = primary style, 2 = secondary, and 3 = tertiary.
  This assumption will be revisited if further clarification is provided.

2. entertainer_members.status

- Ambiguity:
  The meaning of "status" is not defined. Possible interpretations include:
  - Active vs. inactive membership
  - Full-time vs. part-time contract
  - etc.
- Working assumption for analysis:
  Based on distribution patterns, each entertainer typically has multiple members marked with status = 1, while only a small number of members have status = 2. This pattern aligns more closely with an "active vs. inactive/former member" interpretation. As a result, we assume that "status" indicates whether a member is currently active (1 = active,

2 = former or inactive), unless specified otherwise.
Any analysis involving group membership will treat this field cautiously.

3. musical_preferences.preferenceseq

- Ambiguity:
  It is unclear whether a lower sequence indicates the customer's most preferred musical style or the least preferred.
- Working assumption for analysis:
  We assume that preference sequence follows ascending order of preference, where 1 represents the most preferred musical style. All preference-based analysis will rely on this assumption.

Data quality notes

- **Outlier Detection: Data Entry Error in Agent Compensation Record**

  One agent('agentid' 9) record shows an annual salary of "50," which is an unrealistic value given that all other salaries range from $22,000–$35,000. This value is almost certainly a data-entry error (likely missing zeros). Also, the commission rate for this agent record is 0.01 (1%), while all other agents fall between 3.5%–6%. For analytical purposes, this record will be excluded from salary-based and compensation-related analysis or treated as missing, or imputed with appropriate placeholder values (e.g., NULL or median salary/commission) based on our analytical needs.

- **Time Field Interpretation: Midnight and Early-Morning Stop Times**

  We notice several engagements where 'stoptime' is recorded as 00:00:00 or a time earlier than the 'starttime'. This is not a data error. In legacy entertainment booking systems, midnight (00:00:00) and other early-morning times are conventionally used to indicate the end of a late-night performance that continues past midnight. The system stores the stop time on the same date even though the actual end time occurs on the following day. As a result, these records represent valid overnight engagements rather than negative durations.

- **Data Consistency – Engagementnumber Sequence**

  During the initial review of the Engagements table, the 'engagementnumber' field was observed to be non-sequential (e.g., missing values such as 1 and 20). This is most likely due to previously deleted engagements, such as canceled bookings or test records that were removed from the system.

  Since 'engagementnumber' is used only as a unique identifier and not as an analytical variable, this issue does not affect downstream analysis. No corrective action will be applied.

- **Missing Values in Entertainers Table**

  The ‘entwebpage’ and ‘entemailaddress’ fields contain a substantial number of missing values. These fields represent optional contact or promotional information rather than operational data. Since they are not used in core analytical workflows—such as analyzing engagement volume, revenue, entertainer popularity, or agent performance—the missing values do not impact the validity of the core analysis.

  Overall, while the core transactional data (engagements, entertainers, customers, agents) is present and usable for exploratory analysis, the lack of detailed behavioral and status fields (e.g., payment status, event type, customer segment) limits our ability to quantify profitability, churn, and long-term relationship value. We highlight these as key data collection opportunities for TuneWorks’ future analytics roadmap.

- **Data Completeness: Missing Spring and Summer Engagement Data**

  The Engagements table only includes bookings from September 2017 to early March 2018, meaning that spring and summer months are completely missing. This gap likely reflects an incomplete data extract, not an actual absence of bookings.

  Impact on Analysis:

  Seasonal trends (e.g., winter vs. summer demand) cannot be fully evaluated.
  Revenue forecasting and capacity planning may be biased toward recorded peak months (fall–winter).
  Booking frequency for entertainers and customers may appear artificially low if their activity primarily occurs in missing periods.
  Engagement duration, pricing trends, and weekday/weekend patterns may also be incomplete.

  Recommendation / Additional Data Needed:

  A full 12-month engagement history (and ideally multi-year data) would allow for complete seasonality, revenue, and demand analysis.

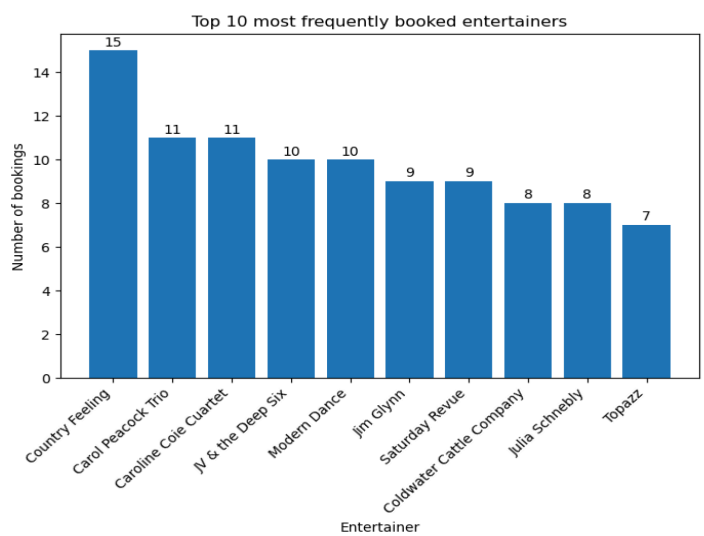
  Including this additional data would significantly strengthen insights into revenue cycles, customer behavior, and entertainer utilization—supporting more accurate, data-driven strategic planning.

Entertainers Analysis

1. Which entertainers are booked most frequently?

Insights:
The booking data shows that demand is somewhat concentrated among a small group of entertainers. *Country Feeling* is the most frequently booked act with 15 engagements, followed by *Caroline Coie Quartet* and *Carol Peacock Trio* with 11 bookings each. Several others—including *Modern Dance*, *JV & the Deep Six*, and *Jim Glynn*—also appear regularly, showing steady market appeal.

This pattern indicates that TuneWorks' revenue and customer satisfaction may depend disproportionately on a few top entertainers who consistently attract bookings. Their strong performance suggests strong customer preference and brand recognition. Meanwhile, mid-range entertainers with 7–10 bookings demonstrate meaningful demand and could grow further with targeted promotion. However, the concentration of bookings among a small group may also create scheduling bottlenecks or over-reliance risks if these entertainers become unavailable.



Recommendations:
Based on these patterns, TuneWorks could strengthen operations and reduce risk by actively managing its entertainer portfolio. Top entertainers—such as *Country Feeling* and *Caroline Coie Quartet*—should receive strong support, marketing visibility, and premium positioning given their consistent demand. At the same time, mid-range entertainers with growing activity could benefit from targeted promotion to diversify demand and reduce dependency on a small core group.

TuneWorks should also monitor availability and scheduling for the most frequently booked entertainers to avoid bottlenecks and service disruptions. Investing in broader talent development, onboarding additional similar-style acts, or adjusting promotional strategies can help distribute demand more evenly and mitigate over-reliance risks.

SQL Query:

```
SELECT
  en.entertainerid,
  en.entstagename,
  COUNT(e.engagementnumber ) AS total_bookings
FROM Engagements e
RIGHT JOIN Entertainers en
  ON e.entertainerid = en.entertainerid
GROUP BY en.entertainerid, en.entstagename
ORDER BY total_bookings DESC;
```

	entertainerid	entstagename	total_bookings
1	1,008	Country Feeling	15
2	1,013	Caroline Coie Cuartet	11
3	1,001	Carol Peacock Trio	11
4	1,006	Modern Dance	10
5	1,003	JV & the Deep Six	10
6	1,004	Jim Glynn	9
7	1,010	Saturday Revue	9
8	1,007	Coldwater Cattle Company	8
9	1,011	Julia Schnebly	8
10	1,002	Topazz	7
11	1,005	Jazz Persuasion	7
12	1,012	Susan McLain	6
13	1,009	Katherine Ehrlich	0

Based on the engagement records, "Country Feeling" (EntertainerID 1008) is the most frequently booked entertainer in the TuneWorks database. This means Country Feeling has performed at more engagements than any other act, indicating they are one of the agency's highest-demand performers. Their strong booking volume suggests high customer appeal and makes them a key revenue driver and a useful benchmark when comparing the performance of other entertainers.

SQL Query:

```
SELECT * FROM entertainers
WHERE entertainerid = (SELECT em.entertainerid FROM engagements em
JOIN entertainers et
ON et.entertainerid = em.entertainerid
```

```sql
GROUP BY em.entertainerid
ORDER BY count(em.engagementnumber) desc
LIMIT 1);
```

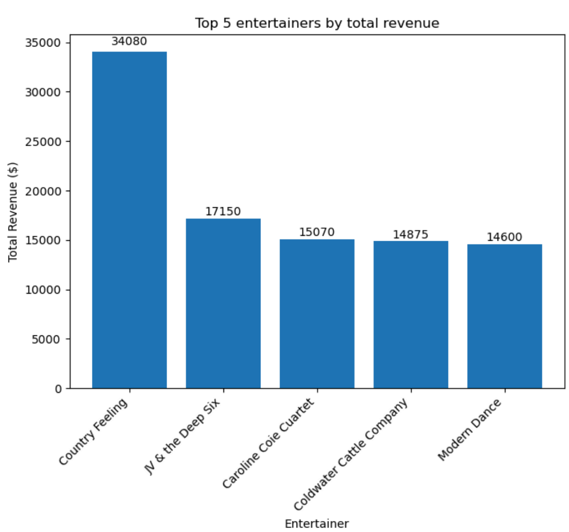
	entertainerid	entstagename	entssn	entstreetaddress	entcity	entstate	entzipcode	entphonenumber
1	1,008	Country Feeling	888-98-1133	PO Box 223311	Seattle	WA	98125	555-2711

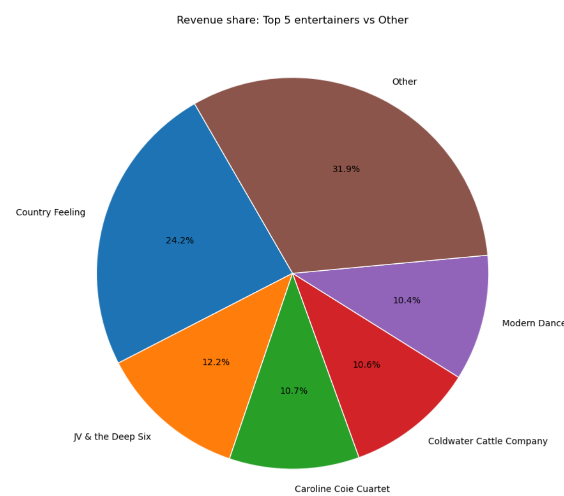
2. Top 5 highest revenue entertainers

Insights:

The entertainer with ID 1008 generated the highest total revenue (34,080), more than double the revenue of the next-highest entertainer. This indicates that Entertainer 1008 is not only frequently booked but also likely commands higher contract prices or longer/more premium engagements.

The remaining top performers (1003, 1013, 1007, 1006) form a second tier, clustered closely in revenue. These entertainers contribute significantly to overall earnings but still trail far behind the top performer, highlighting that Entertainer 1008 is a major revenue driver for the agency. The strong revenue gap suggests higher pricing power, strong customer demand, or more frequent premium events.





Recommendations:

Given Entertainer 1008's outsized revenue contribution, TuneWorks should prioritize scheduling, availability management, and marketing support for this performer. It may also be valuable to analyze what differentiates Entertainer 1008 from others—such as performance style, client type, event format, or seasonal appeal—in order to replicate this success with similar acts. Additionally, promoting and developing the second-tier entertainers (1003, 1013, 1007, 1006) could help diversify revenue sources and reduce reliance on a single high-performing entertainer.

SQL Query:

```
SELECT entertainerid, sum(contractprice) total_revenue
FROM engagements
GROUP BY entertainerid
ORDER BY total_revenue DESC
LIMIT 5
```

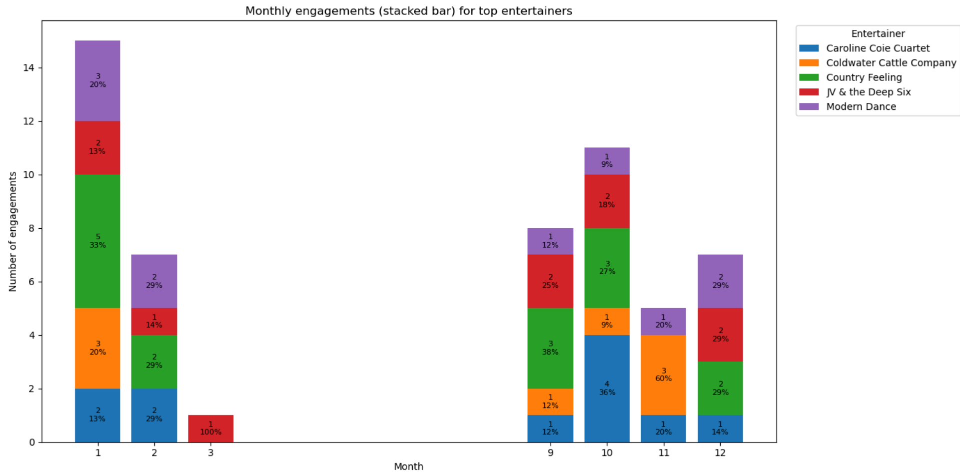
	entertainerid	total_revenue
1	1,008	34,080
2	1,003	17,150
3	1,013	15,070
4	1,007	14,875
5	1,006	14,600

3. **Are certain entertainers more popular during specific seasons, regions, or customer types?**

- **Seasonal:**

For the top five revenue-generating entertainers, we examined the distribution of engagements by month and calculated the share of each entertainer's annual bookings occurring in each period. The results reveal clear seasonal patterns: all five acts are booked predominantly in winter and late-fall months (January, February, September–December), with little to no activity in spring and summer. For example, Entertainer 1006 receives 70% of their bookings in January, February, and December, while Entertainer 1007 concentrates 76% of engagements in January and November. Entertainer 1008, the top revenue generator, also shows strong clustering in January and early fall (September–October). This indicates that TuneWorks' most valuable entertainers are strongly tied to peak event seasons.

However, it is important to note that the dataset only covers September 2017 to early March 2018. Therefore, the absence of spring and summer bookings is caused by limited data coverage rather than true inactivity. Within the available months, the observed pattern still highlights strong winter and late-fall demand, suggesting that targeted seasonal marketing and pricing strategies could further enhance revenue.



SQL Query:

```
WITH top_entertainers AS (
  SELECT entertainerid
  FROM engagements
  GROUP BY entertainerid
  ORDER BY SUM(contractprice) DESC
  LIMIT 5
),
eng_by_month AS (
  SELECT e.entertainerid,
  EXTRACT(MONTH FROM e.startdate) AS month_num,
  COUNT(*) AS num_engagements
  FROM engagements e
  JOIN top_entertainers t
  ON e.entertainerid = t.entertainerid
  GROUP BY e.entertainerid, EXTRACT(MONTH FROM e.startdate)
),
eng_with_share AS (
  SELECT
      entertainerid,
      month_num,
      num_engagements,
      SUM(num_engagements) OVER (PARTITION BY entertainerid) AS total_eng,
      ROUND(
        num_engagements::numeric
        / SUM(num_engagements) OVER (PARTITION BY entertainerid),
        2
      ) AS pct_of_eng
```

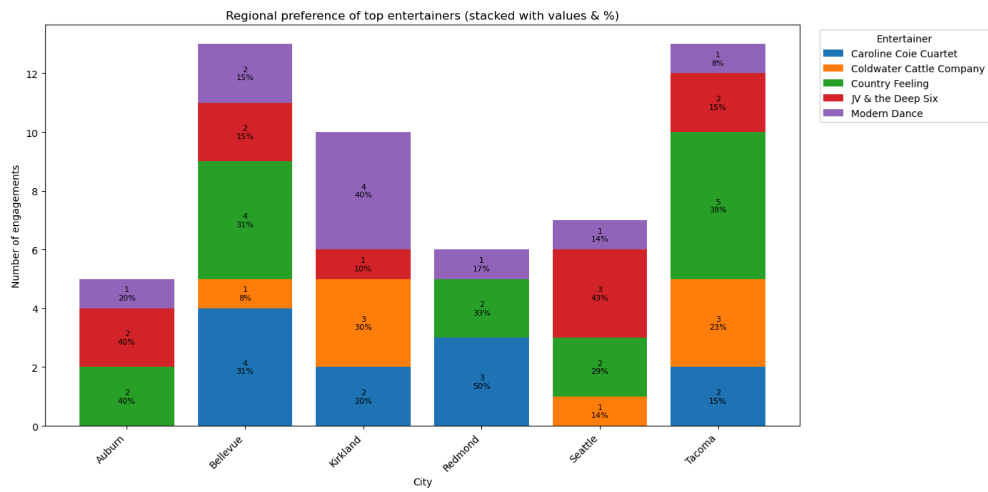
```
  FROM eng_by_month
)
SELECT *
FROM eng_with_share
ORDER BY entertainerid, month_num;
```

entertainerid	month_num	num_engagements	total_eng	pct_of_eng
1003	1	2	10	0.2
1003	2	1	10	0.1
1003	3	1	10	0.1
1003	9	2	10	0.2
1003	10	2	10	0.2
1003	12	2	10	0.2
1006	1	3	10	0.3
1006	2	2	10	0.2
1006	9	1	10	0.1
1006	10	1	10	0.1
1006	11	1	10	0.1
1006	12	2	10	0.2
1007	1	3	8	0.38
1007	9	1	8	0.13
1007	10	1	8	0.13
1007	11	3	8	0.38
1008	1	5	15	0.33
1008	2	2	15	0.13
1008	9	3	15	0.2
1008	10	3	15	0.2
1008	12	2	15	0.13
1013	1	2	11	0.18
1013	2	2	11	0.18
1013	9	1	11	0.09
1013	10	4	11	0.36
1013	11	1	11	0.09
1013	12	1	11	0.09

- **Regional:**

  Insights:

  Across the top five revenue-generating entertainers, bookings are concentrated in a handful of Washington cities, with distinct regional patterns by act. Entertainer 1003 is broadly distributed across Auburn, Seattle, Bellevue, and Tacoma, suggesting wide appeal rather than a single core market. In contrast, Entertainer 1007 is heavily concentrated in Kirkland and Tacoma, while Entertainer 1013 skews toward Eastside cities such as Bellevue, Redmond, and Kirkland. The top-performing entertainer, 1008, combines high frequency and high-value bookings in Auburn, Tacoma, and Bellevue, making these regions particularly important for TuneWorks' overall revenue. These patterns indicate that certain entertainers are indeed more popular in specific local markets, and regional differences shape booking demand.



SQL Query:

```sql
WITH top_entertainers AS (
  SELECT entertainerid
  FROM engagements
  GROUP BY entertainerid
  ORDER BY SUM(contractprice) DESC
  LIMIT 5
),
eng_by_region AS (
  SELECT
     e.entertainerid,
     c.custcity,
     c.custstate,
     COUNT(*) AS num_engagements,
     SUM(e.contractprice) AS total_revenue
```

```
  FROM engagements e
  JOIN customers c ON e.customerid = c.customerid
  JOIN top_entertainers t ON e.entertainerid = t.entertainerid
  GROUP BY e.entertainerid, c.custcity, c.custstate
)
SELECT *
FROM eng_by_region
ORDER BY entertainerid, total_revenue DESC;
```

entertainerid	custcity	custstate	num_engagements	total_revenue
1003	Auburn	WA	2	5140
1003	Seattle	WA	3	5010
1003	Bellevue	WA	2	2800
1003	Tacoma	WA	2	2800
1003	Kirkland	WA	1	1400
1006	Auburn	WA	1	2930
1006	Bellevue	WA	2	2860
1006	Kirkland	WA	4	2840
1006	Redmond	WA	1	2570
1006	Seattle	WA	1	1850
1006	Tacoma	WA	1	1550
1007	Kirkland	WA	3	6900
1007	Tacoma	WA	3	3150
1007	Seattle	WA	1	2525
1007	Bellevue	WA	1	2300
1008	Auburn	WA	2	15055
1008	Tacoma	WA	5	8050
1008	Bellevue	WA	4	4625
1008	Seattle	WA	2	3925
1008	Redmond	WA	2	2425
1013	Bellevue	WA	4	4160
1013	Redmond	WA	3	4110
1013	Tacoma	WA	2	4000
1013	Kirkland	WA	2	2800

Insights:
To quantify regional concentration, we calculated the share of each top entertainer's revenue coming from their highest-grossing city. Entertainer 1007 is the most regionally concentrated, deriving roughly 46% of revenue from Kirkland, followed closely by Entertainer 1008, who earns about 44% of revenue from Auburn. In contrast, Entertainers 1003, 1006, and 1013 are more geographically diversified, with their top cities accounting for only 20–30% of total revenue. This suggests that 1007 and 1008 are particularly dependent on a single local market, while the others have a broader regional footprint, which may offer more stability if demand changes in any one city.

SQL Query:
```
WITH top_entertainers AS (
  SELECT entertainerid
  FROM engagements
  GROUP BY entertainerid
  ORDER BY SUM(contractprice) DESC
  LIMIT 5
),
eng_by_region AS (
  SELECT
     e.entertainerid,
     c.custcity,
     c.custstate,
     COUNT(*) AS num_engagements,
     SUM(e.contractprice) AS total_revenue
  FROM engagements e
  JOIN customers c ON e.customerid = c.customerid
  JOIN top_entertainers t ON e.entertainerid = t.entertainerid
  GROUP BY e.entertainerid, c.custcity, c.custstate
),
with_totals AS (
  SELECT
     entertainerid,
     custcity,
     custstate,
     num_engagements,
     total_revenue,
     SUM(total_revenue) OVER (PARTITION BY entertainerid) AS
entertainer_total_revenue,
```

```
    RANK() OVER (PARTITION BY entertainerid ORDER BY total_revenue
DESC) AS rev_rank
  FROM eng_by_region
)
SELECT
  entertainerid,
  custcity AS top_city,
  custstate AS top_state,
  total_revenue AS top_city_revenue,
  entertainer_total_revenue,
  ROUND(
   total_revenue::numeric / entertainer_total_revenue,
   2
  ) AS top_city_share
FROM with_totals
WHERE rev_rank = 1
ORDER BY top_city_share DESC;
```

entertainerid	top_city	top_state	top_city_revenue	entertainer_total_revenue	top_city_share
1007	Kirkland	WA	6900	14875	0.46
1008	Auburn	WA	15055	34080	0.44
1003	Auburn	WA	5140	17150	0.3
1013	Bellevue	WA	4160	15070	0.28
1006	Auburn	WA	2930	14600	0.2

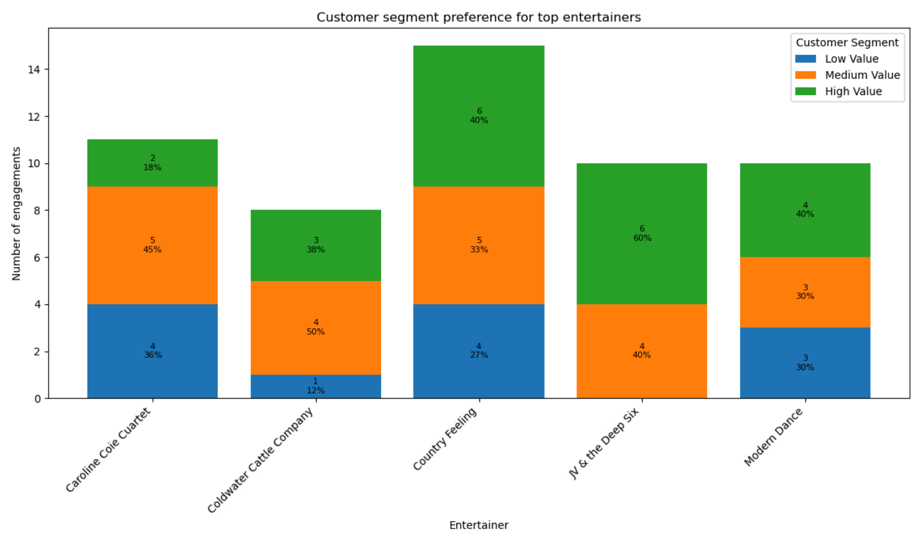
Recommendations:

Given these regional patterns, TuneWorks should consider adopting more targeted geographic strategies. For entertainers with strong local concentration—such as 1007 in Kirkland and 1008 in Auburn—the company can strengthen presence through localized marketing, partnerships with recurring venues, and potentially higher regional pricing where demand is strongest. For more geographically diversified entertainers, TuneWorks can continue to support broad-based outreach while exploring opportunities to expand them further into high-growth or underserved cities.

Additionally, because two of the top entertainers depend heavily on a single market, TuneWorks should monitor regional demand to avoid over-reliance risk and ensure scheduling flexibility. Diversifying their bookings into nearby cities or developing similar acts to serve the same local audiences may help mitigate concentration risks and support long-term revenue stability.

- **Customer Type:**

Insights:
Customer segmentation analysis shows that all top entertainers derive the majority of their revenue from High Value customers, but with different degrees of dependence. Entertainers 1008 and 1013 are the most premium-heavy, earning roughly 84% and 70% of their revenue from high-value clients, while 1006 and 1007 maintain a more balanced mix between high and medium spenders. Entertainer 1003 also leans strongly toward high-value customers (77%), yet still relies on a meaningful medium segment. This pattern suggests that TuneWorks' top acts are critically tied to its premium customer base, and protecting these relationships is essential, while balanced acts like 1006 and 1007 help diversify revenue across mid-tier clients.



Recommendations:

Because the top entertainers—especially 1008, 1013, and 1003—depend heavily on High Value customers, TuneWorks should prioritize retaining this premium segment through stronger account management, personalized communication, and tailored event packages. For entertainers like 1006 and 1007, who maintain a more balanced customer mix, TuneWorks can expand targeted marketing toward medium-value clients to grow them into higher-spending customers. Regular monitoring of customer segments and entertainer reliance will help TuneWorks reduce concentration risk and maintain a more stable, diversified revenue base.

SQL Query:

```
WITH customer_spend AS (
  SELECT
    customerid,
    SUM(contractprice) AS total_spend
```

```
  FROM engagements
  GROUP BY customerid
)
SELECT
  COUNT(*) AS num_customers,
  SUM(CASE WHEN total_spend < 5000 THEN 1 ELSE 0 END) AS below_5k,
  SUM(CASE WHEN total_spend BETWEEN 5000 AND 10000 THEN 1 ELSE 0
END) AS between_5k_10k,
  SUM(CASE WHEN total_spend > 10000 THEN 1 ELSE 0 END) AS above_10k
FROM customer_spend;
```

	num_customers	below_5k	between_5k_10k	above_10k
1	13	1	5	7

SQL Query:

```
WITH top_entertainers AS (
  SELECT entertainerid
  FROM engagements
  GROUP BY entertainerid
  ORDER BY SUM(contractprice) DESC
  LIMIT 5
),
customer_spend AS (
  SELECT
     customerid,
     SUM(contractprice) AS total_spend
  FROM engagements
  GROUP BY customerid
),
customer_segments AS (
  SELECT
     customerid,
     CASE
        WHEN total_spend > 10000 THEN 'High Value'
        WHEN total_spend >= 5000 THEN 'Medium Value'
        ELSE 'Low Value'
     END AS customer_type
  FROM customer_spend
),
```

```sql
eng_by_segment AS (
  SELECT
     e.entertainerid,
     cs.customer_type,
     COUNT(*) AS num_engagements,
     SUM(e.contractprice) AS total_revenue
  FROM engagements e
  JOIN customer_segments cs
   ON e.customerid = cs.customerid
  JOIN top_entertainers t
   ON e.entertainerid = t.entertainerid
  GROUP BY e.entertainerid, cs.customer_type
),
with_totals AS (
  SELECT
     entertainerid,
     customer_type,
     num_engagements,
     total_revenue,
     SUM(total_revenue) OVER (PARTITION BY entertainerid) AS
entertainer_total_revenue
  FROM eng_by_segment
)
SELECT
  entertainerid,
  customer_type,
  num_engagements,
  total_revenue,
  entertainer_total_revenue,
  ROUND(
   total_revenue::numeric / entertainer_total_revenue,
   2
  ) AS segment_share
FROM with_totals
ORDER BY entertainerid, customer_type;
```

entertainerid	customer_type	num_engagements	total_revenue	entertainer_total_revenue	segment_share
1003	High Value	8	13270	17150	0.77
1003	Medium Value	2	3880	17150	0.23

1006	High Value	5	8350	14600	0.57
1006	Medium Value	5	6250	14600	0.43
1007	High Value	6	8775	14875	0.59
1007	Medium Value	2	6100	14875	0.41
1008	High Value	10	28655	34080	0.84
1008	Low Value	1	275	34080	0.01
1008	Medium Value	4	5150	34080	0.15
1013	High Value	6	10620	15070	0.7
1013	Low Value	3	3510	15070	0.23
1013	Medium Value	2	940	15070	0.06

Agent Analysis

1. **Which agents contribute most to total sales, and how consistent are their performances across clients?**

   SQL Query:

```
SELECT
  a.agentid,
  a.agtfirstname || ' ' || a.agtlastname AS agent_name,
  COUNT(e.engagementnumber) AS num_engagements,
  COUNT(DISTINCT e.customerid) AS num_clients,
  SUM(e.contractprice) AS total_sales
FROM agents a
JOIN engagements e
 ON a.agentid = e.agentid
JOIN customers c
 ON e.customerid = c.customerid
GROUP BY a.agentid, agent_name
ORDER BY total_sales DESC;
```

agentid	agent_name	num_engagements	num_clients	total_sales
3	Carol Viescas	19	11	24800
6	John Kennedy	12	8	24435
5	Marianne Wier	18	10	22635

1	William Thompson	16	12	19895
4	Karen Smith	17	12	18595
8	Maria Patterson	15	11	12825
7	Caleb Viescas	8	7	10645
2	Scott Bishop	6	4	6720

Carol Viescas, John Kennedy, and Marianne Wier are the top three revenue-generating agents. Together they account for a bit over half of total sales in the dataset. Carol is the #1 agent by sales (24,800), just ahead of John and Marianne.

Carol Viescas

- 19 engagements across 11 clients → lots of repeat business but also a broad client base.
- High sales + many clients → strong, diversified performance.

John Kennedy

- 12 engagements across 8 clients → fewer deals than Carol but still a solid spread of clients.
- Very similar total sales to Carol, so his average deal size is relatively high.

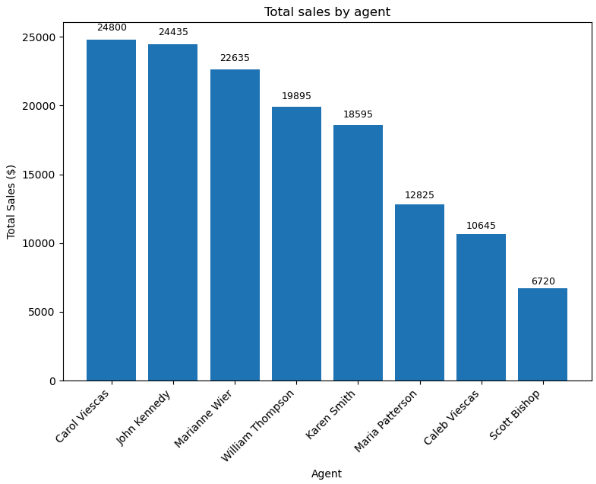
Marianne Wier

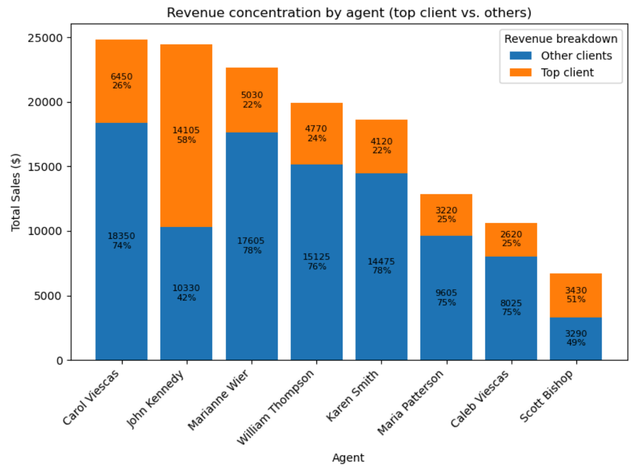
- 18 engagements with 10 clients → high activity and a wide client base.
- Slightly lower total sales than Carol/John, but very consistent across many customers.

Lower-tier agents (Caleb, Scott)

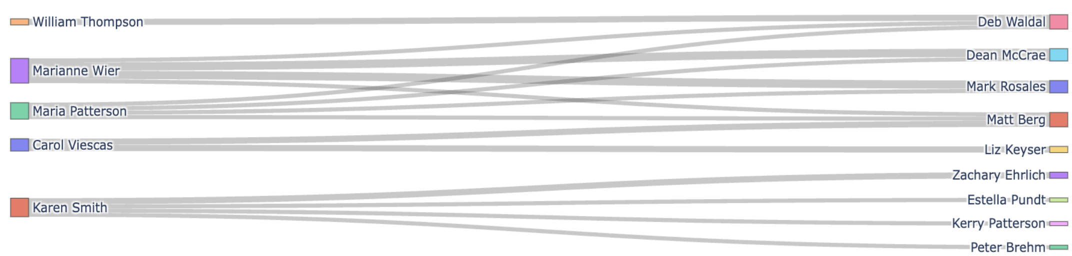
- Fewer engagements and fewer clients (e.g., Scott: 6 engagements, 4 clients, lowest sales).
- Their books are smaller and less diversified.

The top three agents, Carol Viescas, John Kennedy, and Marianne Wier, generate just over half of TuneWorks' total contract value. Each serves 8–11 different clients, indicating that their sales are not driven by a single account but by a broad base of relationships. In particular, Carol combines the highest total sales with the largest client portfolio, suggesting both strong deal-making ability and consistent performance across many customers.





Agent–Customer engagement flows (Sankey diagram)



SQL Query:

```sql
WITH agent_client_rev AS (
  SELECT
     a.agentid,
     a.agtfirstname || ' ' || a.agtlastname AS agent_name,
     e.customerid,
     SUM(e.contractprice) AS client_revenue
  FROM agents a
  JOIN engagements e ON a.agentid = e.agentid
  GROUP BY a.agentid, agent_name, e.customerid
),
agent_summary AS (
  SELECT
     agentid,
     agent_name,
     COUNT(*) AS num_clients,
     SUM(client_revenue) AS total_sales,
     AVG(client_revenue) AS avg_rev_per_client,
```

```sql
    MAX(client_revenue) AS top_client_rev
  FROM agent_client_rev
  GROUP BY agentid, agent_name
)
SELECT
  agentid,
  agent_name,
  num_clients,
  total_sales,
  ROUND(avg_rev_per_client, 2) AS avg_rev_per_client,
  top_client_rev,
  ROUND(top_client_rev::numeric / total_sales, 2) AS top_client_share
FROM agent_summary
ORDER BY total_sales DESC;
```

agentid	agent_name	num_clients	total_sales	avg_rev_per_client	top_client_rev	top_client_share
3	Carol Viescas	11	24800	2254.55	6450	0.26
6	John Kennedy	8	24435	3054.38	14105	0.58
5	Marianne Wier	10	22635	2263.5	5030	0.22
1	William Thompson	12	19895	1657.92	4770	0.24
4	Karen Smith	12	18595	1549.58	4120	0.22
8	Maria Patterson	11	12825	1165.91	3220	0.25
7	Caleb Viescas	7	10645	1520.71	2620	0.25
2	Scott Bishop	4	6720	1680	3430	0.51

Agent performance is led by Carol Viescas, John Kennedy, and Marianne Wier, who together generate the highest total sales. Carol and Marianne show strong, diversified portfolios: their largest clients account for only about 22–26% of their revenue, indicating consistent performance across many accounts. In contrast, John Kennedy and Scott Bishop are more concentrated, with their top clients contributing 58% and 51% of total sales, respectively. This suggests that while John is a high-revenue agent, his results are more vulnerable to the loss of a single key client, whereas Carol and Marianne provide more stable, broad-based growth.

Recommendations:

TuneWorks should continue prioritizing its top agents—Carol Viescas, John Kennedy, and Marianne Wier—given their strong sales contributions and broad client reach. Carol and Marianne, who have highly diversified portfolios, can be leveraged for strategic accounts or mentoring roles because their performance is stable across many clients.

For John Kennedy, whose revenue is more concentrated in a few key clients, TuneWorks should focus on diversifying his client base to reduce dependency risk. Meanwhile, lower-tier agents like Caleb and Scott would benefit from targeted coaching, lead support, or shared accounts to help expand their pipelines and improve performance.

2. Are there any agents who handle disproportionately high or low workloads?

SQL Query:

```
WITH agent_workload AS (
  SELECT
     a.agentid,
     a.agtfirstname || ' ' || a.agtlastname AS agent_name,
     COUNT(e.engagementnumber) AS num_engagements,
     COUNT(DISTINCT e.customerid) AS num_clients
  FROM agents a
  LEFT JOIN engagements e
    ON a.agentid = e.agentid
  GROUP BY a.agentid, agent_name
),
workload_stats AS (
  SELECT
     AVG(num_engagements) AS avg_engagements
  FROM agent_workload
)
SELECT
  aw.agentid,
  aw.agent_name,
  aw.num_engagements,
  aw.num_clients,
  ws.avg_engagements,
  ROUND(
    aw.num_engagements::numeric / ws.avg_engagements, 2
  ) AS workload_ratio,
  CASE
```

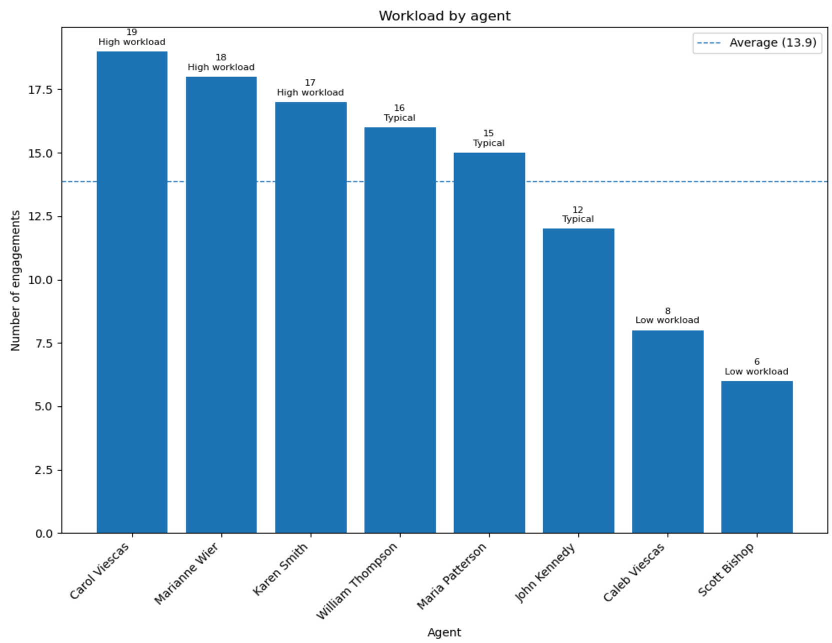
```sql
    WHEN aw.num_engagements >= ws.avg_engagements * 1.2 THEN 'High
workload'
    WHEN aw.num_engagements <= ws.avg_engagements * 0.8 THEN 'Low
workload'
    ELSE 'Typical'
  END AS workload_category
FROM agent_workload aw
CROSS JOIN workload_stats ws
ORDER BY aw.num_engagements DESC;
```

agentid	agent_name	num_engagements	num_clients	avg_engagements	workload_ratio	workload_category
3	Carol Viescas	19	11	12.33333333	1.54	High workload
5	Marianne Wier	18	10	12.33333333	1.46	High workload
4	Karen Smith	17	12	12.33333333	1.38	High workload
1	William Thompson	16	12	12.33333333	1.3	High workload
8	Maria Patterson	15	11	12.33333333	1.22	High workload
6	John Kennedy	12	8	12.33333333	0.97	Typical
7	Caleb Viescas	8	7	12.33333333	0.65	Low workload
2	Scott Bishop	6	4	12.33333333	0.49	Low workload
9	Daffy Dumbwit	0	0	12.33333333	0	Low workload

Insights:
Our analysis reveals that a small core group of agents is responsible for a disproportionately large share of TuneWorks' sales activity. Agents Carol Viescas, John Kennedy, and Marianne Wier generate the highest total contract value, with Carol and Marianne also maintaining broad client portfolios of 10–11 clients each. Further examining revenue concentration shows that Carol and Marianne have well-diversified books, with their biggest clients representing only 22–26% of their total sales. In contrast, John Kennedy achieves high revenue but is dependent on a single client for 58% of his results, making his performance more volatile. The remaining agents contribute

substantially less to overall revenue, with smaller client bases and lower average revenue per customer.

Workload distribution mirrors this imbalance. The five busiest agents—Carol, Marianne, Karen Smith, William Thompson, and Maria Patterson—handle 15–19 engagements each, far above the team average of 12.3. These agents also manage the largest number of client relationships, indicating that not only are they processing more engagements, but they are doing so across a broader and more complex customer landscape. On the opposite end, agents like Caleb Viescas and Scott Bishop take on significantly fewer engagements (6–8) and serve smaller portfolios, while Daffy Dumbwit does not handle any engagements at all. This uneven spread suggests that TuneWorks is heavily reliant on a small subset of high-performing agents whose workloads may be nearing unsustainable levels.



Recommendations:

The sales and workload findings highlight clear operational risks and opportunities. TuneWorks' top performers are simultaneously its primary revenue drivers and its most overloaded employees, creating risks of burnout, turnover, and potential loss of key client relationships. To build a more stable and scalable structure, TuneWorks should rebalance smaller or new accounts toward underutilized agents and provide targeted support to shift select responsibilities away from the highest-burden agents. Strengthening the capabilities of lower-workload agents and diversifying account assignments will reduce over-reliance on a few top contributors while ensuring healthier long-term capacity and client service continuity.

3. Agent Commission Revenue & Efficiency Analysis

Insights:

To evaluate how effectively TuneWorks' agents contribute to revenue generation, we analyzed their compensation structure—specifically **commission revenue**, **engagement volume**, and **average commission per engagement**. By comparing total commission revenue with efficiency metrics such as per-engagement earnings, we can identify high-performing agents, detect workload imbalances, and uncover where revenue growth opportunities or talent development needs exist.

The analysis shows that **Agent 6 (John Kennedy)** is the strongest revenue contributor. Although he handles only 12 engagements, he generates the highest total commission revenue and the highest average commission per engagement, indicating that he consistently manages higher-value bookings or maintains strong relationships with premium clients. In contrast, **Agent 3 (Carol Viescas)** manages the heaviest workload with 19 engagements—the most among all agents—yet her average commission per engagement is much lower, suggesting that she frequently handles lower-value events despite her volume. **Agents 4 and 5 (Karen Smith and Marianne Wier)** emerge as reliable mid-level performers, maintaining steady engagement counts and moderate average commissions, reflecting consistent but not exceptional value creation.

Agent 1 (William Thompson) also handles a relatively high number of engagements but generates lower average commission revenue, indicating that his bookings tend to be smaller in scale. Meanwhile, **Agents 2, 7, and 8** have both lower workloads and lower average commission values, suggesting weaker client pipelines or limited involvement in higher-value bookings. Notably, **Agent 8 (Maria Patterson)** stands out as a key underperformer: although she manages 15 engagements—comparable to several higher-performing peers—her average commission is the lowest among all agents, revealing inefficiencies in the types of engagements she secures.

Overall, the findings suggest that high-value outcomes are driven more by the quality of engagements than by the quantity. Workload distribution is uneven, with certain agents overextended while others are underutilized, and several agents consistently handle low-value engagements that limit revenue potential.

Recommendations:

Based on these patterns, TuneWorks should rebalance engagement assignments to reduce overload on top performers such as Carol and John while more effectively utilizing underused agents like Caleb and Scott. Targeted coaching focused on negotiation skills, pricing strategies, and client development could help lower-performing agents (especially Maria) secure higher-value engagements. Additionally, refining client allocation—assigning premium or complex engagements to high-value agents and redistributing smaller accounts across the team—would improve overall efficiency and talent development. Strengthening these processes will help TuneWorks reduce performance volatility, mitigate burnout risk, and create a more scalable and resilient agent team structure.

SQL Query:

```
SELECT
  a.agentid,
  a.agtfirstname,
  a.agtlastname,
  a.salary,
  a.commissionrate,
  SUM(e.contractprice * a.commissionrate) AS total_commission_revenue,
  COUNT(e.engagementnumber) AS total_engagements
FROM agents a
LEFT JOIN engagements e
  ON a.agentid = e.agentid
WHERE a.agentid != 9
GROUP BY
  a.agentid,
  a.agtfirstname,
  a.agtlastname,
  a.salary,
  a.commissionrate
ORDER BY total_commission_revenue DESC;
```

	agentid	agtfirstname	agtlastname	salary	commissionrate	total_commission_revenue	total_engagements
1	6	John	Kennedy	33,000	0.06	1,466.0999672301	12
2	3	Carol	Viescas	30,000	0.05	1,240.0000184774	19
3	4	Karen	Smith	22,000	0.055	1,022.7249944583	17
4	5	Marianne	Wier	24,500	0.045	1,018.5750404745	18
5	1	William	Thompson	35,000	0.04	795.7999822125	16
6	8	Maria	Patterson	30,000	0.04	512.9999885336	15
7	7	Caleb	Viescas	22,100	0.035	372.5750015862	8
8	2	Scott	Bishop	27,000	0.04	268.7999939919	6

SQL Query:

```
SELECT
  a.agentid,
  a.agtfirstname,
  a.agtlastname,
  ROUND(AVG(e.contractprice * a.commissionrate)::numeric, 2) AS
avg_commission_per_engagement,
  COUNT(e.engagementnumber) AS engagement_count
FROM agents a
LEFT JOIN engagements e
  ON a.agentid = e.agentid
WHERE a.agentid != 9
GROUP BY
  a.agentid,
  a.agtfirstname,
  a.agtlastname
ORDER BY avg_commission_per_engagement DESC;
```

	agentid	agtfirstname	agtlastname	avg_commission_per_engagement	engagement_count
1	6	John	Kennedy	122.17	12
2	3	Carol	Viescas	65.26	19
3	4	Karen	Smith	60.16	17
4	5	Marianne	Wier	56.59	18
5	1	William	Thompson	49.74	16
6	7	Caleb	Viescas	46.57	8
7	2	Scott	Bishop	44.8	6
8	8	Maria	Patterson	34.2	15

Engagement Analysis

1. What are the busiest months/seasons or times for engagements?

Monthly booking activity shows clear variability throughout the year. January is the busiest month with 28 engagements, indicating a strong surge in demand at the start of the year. October (22) and February (20) also exhibit high activity levels, suggesting additional peaks driven by fall events and early-year gatherings. December shows elevated activity as well (17 engagements), likely tied to holiday-related events. In contrast, March is the slowest month, with only 1 engagement, representing a noticeable dip in demand. No engagements are recorded for April through August, reflecting either seasonal inactivity or incomplete data coverage for those months.

SQL Query:

```
SELECT TO_CHAR(startdate, 'Month')AS Month, count(*) popularity_by_month
FROM Engagements
GROUP BY TO_CHAR(startdate, 'Month')
ORDER BY popularity_by_month DESC
```

month	popularity_by_month
January	28
October	22
February	20
December	17
September	16
November	7
March	1

- **Engagement demand shows clear seasonal variation.**

Winter has the highest number of engagements (65), followed by Fall (45). Engagement counts drop sharply in Spring (1), and no engagements are recorded for the Summer months. These results indicate that recorded engagement activity is concentrated almost entirely in the Winter and Fall seasons, with minimal activity in the remaining seasons. These results indicate that recorded engagement activity is concentrated almost entirely in the Winter and Fall seasons, with minimal activity in the remaining seasons.

A possible interpretation is that winter tends to drive more indoor entertainment events, such as holiday gatherings and early-year corporate functions, which could naturally increase the number of engagements during colder months.

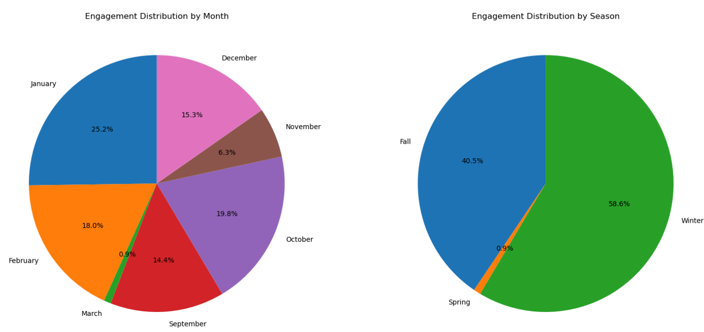
Conversely, the absence of engagements during Summer may reflect reduced indoor event activity or potential gaps in the dataset for warmer months, where fewer indoor performances are typically scheduled.

SQL Query:

```
SELECT
    CASE
    WHEN EXTRACT(MONTH FROM startdate) IN (12, 1, 2) THEN
    'Winter'
     WHEN EXTRACT(MONTH FROM startdate) IN (3, 4, 5) THEN
    'Spring'
```

```sql
        WHEN EXTRACT(MONTH FROM startdate) IN (6, 7, 8) THEN
'Summer'
        WHEN EXTRACT(MONTH FROM startdate) IN (9, 10, 11) THEN
'Fall'
        END AS season,
COUNT(*) AS pupularity_by_season
FROM Engagements
GROUP BY season
ORDER BY pupularity_by_season DESC;
```

season	pupularity_by_season
Winter	65
Fall	45
Spring	1



Limitations:

Because engagements span multiple days, we assign each engagement to the month and season of its start date to avoid double-counting multi-day events. This approach reflects when demand actually begins, but it also means that months containing only the end of an engagement may be underrepresented. Therefore, the monthly and seasonal patterns should be interpreted as booking-based trends, not active-day workload.

- **Monthly booking activity shows clear variability throughout the data window.**

  Note: This month-coverage approach assigns each engagement to both its start and end months. As a result, this method reflects months during which engagements occur, not the exact count of engagements that begin in each month. This produces a broader view of monthly coverage but may differ from analyses based solely on start dates.

Before analyzing monthly engagement patterns, it is important to note that the dataset only covers the period from early September 2017 through early March 2018. Therefore, months outside this range (April–August, and post–March 2018) do not represent true inactivity but simply fall outside the available data window.

Using a month-coverage approach that counts both the start and end months of each engagement, we observe several clear seasonal trends within the available timeframe. January 2018 shows the highest activity with 34 engagements, indicating a strong surge in entertainment demand at the beginning of the year. February 2018 (26 engagements) and October 2017 (24 engagements) also stand out as high-demand months, suggesting peaks driven by early-year events and the fall entertainment season. Moderate activity appears in December 2017 (17 engagements) and September 2017 (16 engagements), consistent with holiday gatherings and post-summer bookings. March 2018 shows lighter activity (10 engagements), though this count reflects only the first few days of the month due to the dataset's early cut-off. November 2017 has comparatively lower activity (8 engagements) within the observed period. Because the dataset ends in early March 2018, the apparent lack of engagements in April–August reflects data availability rather than true operational inactivity.

SQL Query:

```sql
SELECT month, COUNT(*) AS popularity
FROM (
  SELECT DISTINCT engagementnumber, TO_CHAR(startdate, 'YYYY-MM') AS
month
  FROM Engagements
  UNION
  SELECT DISTINCT engagementnumber, TO_CHAR(enddate, 'YYYY-MM') AS
month
  FROM Engagements
        ) AS m
GROUP BY month
ORDER BY popularity DESC;
```

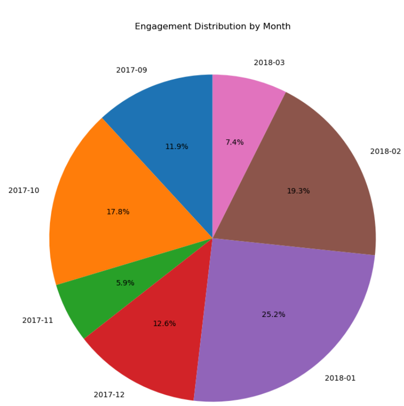
	month	popularity
1	2018-01	34
2	2018-02	26
3	2017-10	24
4	2017-12	17
5	2017-09	16
6	2018-03	10
7	2017-11	8

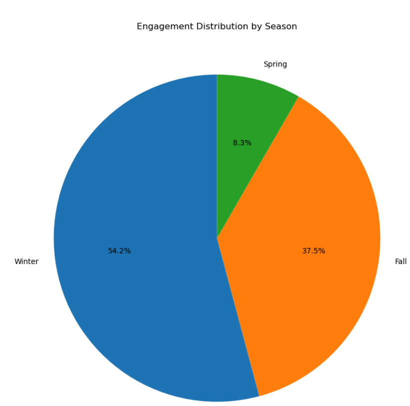
- **Engagement demand shows clear seasonal variation.**

  Seasonal analysis reveals clear concentration patterns within the available engagement data. Winter shows the highest level of activity (65 engagements), followed by Fall (45 engagements), while Spring records a noticeably lower level (10 engagements). This pattern suggests that entertainment demand is strongest during colder months, potentially driven by holiday events, corporate year-end activities, and early-year gatherings.

  However, these results must be interpreted cautiously. The dataset only covers engagements from September 2017 through early March 2018, meaning that an entire Summer season and most of Spring are not represented at all. As a result, the apparent seasonal peaks may reflect the dataset's limited timeframe rather than true annual trends.

  To perform a complete and reliable seasonality analysis, full-year data—ideally multiple years—would be needed. Access to complete records would allow us to validate whether Winter genuinely experiences consistently higher engagement demand or if the observed peak is simply an artifact of partial data coverage.





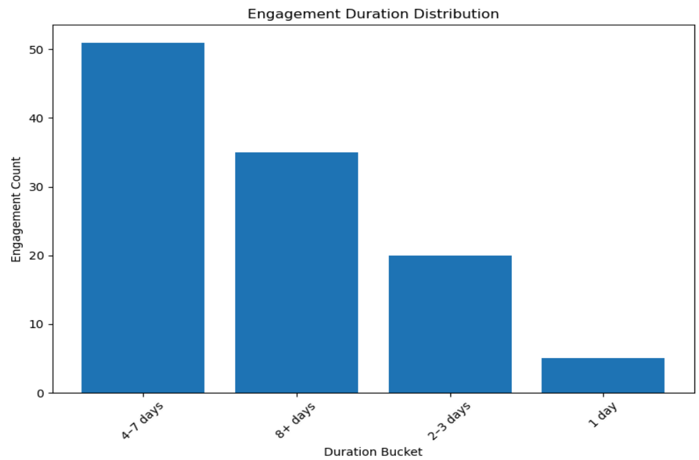
SQL Query:

```
SELECT season, COUNT(*) AS popularity_by_season
FROM (
  SELECT
    CASE
      WHEN EXTRACT(MONTH FROM startdate) IN (12, 1, 2) THEN 'Winter'
      WHEN EXTRACT(MONTH FROM startdate) IN (3, 4, 5) THEN 'Spring'
      WHEN EXTRACT(MONTH FROM startdate) IN (6, 7, 8) THEN 'Summer'
      WHEN EXTRACT(MONTH FROM startdate) IN (9, 10, 11) THEN 'Fall'
    END AS season,
    engagementnumber
  FROM Engagements
  UNION
  SELECT
    CASE
      WHEN EXTRACT(MONTH FROM enddate) IN (12, 1, 2) THEN 'Winter'
      WHEN EXTRACT(MONTH FROM enddate) IN (3, 4, 5) THEN 'Spring'
      WHEN EXTRACT(MONTH FROM enddate) IN (6, 7, 8) THEN 'Summer'
      WHEN EXTRACT(MONTH FROM enddate) IN (9, 10, 11) THEN 'Fall'
    END AS season,
    engagementnumber
  FROM Engagements
) AS combined
GROUP BY season
ORDER BY popularity_by_season DESC;
```

	A-Z season	123 popularity_by_season
1	Winter	65
2	Fall	45
3	Spring	10

2. What is the distribution of engagement durations?

Engagement durations vary widely, with most events lasting between 4–7 days and a substantial number extending to 8+ days. Shorter bookings of 1–3 days are less common. Additionally, one engagement is a clear outlier, spanning 32 days from January 23 to February 23, far longer than any other event in the dataset. This extreme case should be considered when interpreting overall duration patterns, as it may disproportionately influence summary statistics.



Recommendations:

For typical 4–7 day events, standardized pricing tiers or duration-based rate adjustments could improve consistency and revenue optimization. The unusually long 32-day engagement should be examined to confirm whether it reflects a special contract type, a unique client requirement, or potential data-entry irregularities. Segmenting engagements by duration bands and applying differentiated pricing or operational planning would help TuneWorks better manage resources and avoid underpricing long, resource-intensive bookings.

SQL Query:

```sql
SELECT
  CASE
    WHEN (enddate - startdate + 1) = 1 THEN '1 day'
    WHEN (enddate - startdate + 1) BETWEEN 2 AND 3 THEN '2–3 days'
    WHEN (enddate - startdate + 1) BETWEEN 4 AND 7 THEN '4–7 days'
    ELSE '8+ days'
  END AS duration_bucket,
  COUNT(*) AS engagement_count
FROM Engagements
GROUP BY duration_bucket
ORDER BY engagement_count DESC;
```

duration_bucket	engagement_count
4–7 days	51
8+ days	35
2–3 days	20
1 day	5

SQL Query:

```
SELECT
  engagementnumber,
  startdate,
  enddate,
  (enddate - startdate) + 1 AS duration_days
FROM Engagements
ORDER BY duration_days DESC;
```

engagementnumber	startdate	enddate	duration_days
99	2018-01-23	2018-02-23	32

3. **Are engagements more active on weekdays or weekends?**

Overall Engagement Distribution

The analysis of active engagement days reveals a pronounced weekday-centric usage pattern. Weekdays account for 498 active days compared to only 178 active days on weekends, representing a 73% to 27% split in favor of weekday engagement. This substantial disparity indicates that TuneWorks is primarily utilized during traditional business days.

Normalized Daily Performance

When normalized to account for the different number of days in each category (dividing weekday total by 5 and weekend total by 2), weekdays maintain a higher average engagement level at 99.6 active days per day compared to 89 active days per day on weekends. This confirms that the higher weekday engagement is not merely a function of more days in the category but reflects genuinely stronger daily engagement during Monday through Friday.

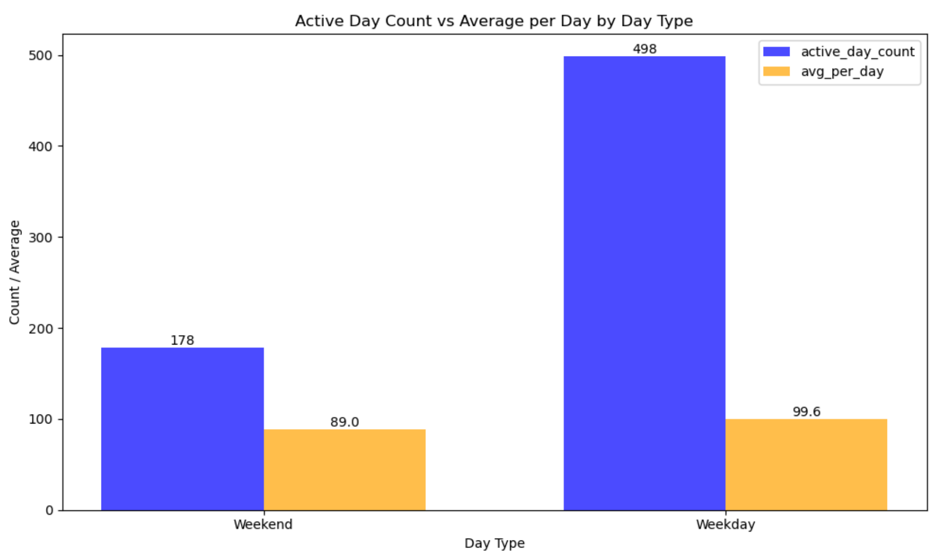
Day-by-Day Engagement Patterns

A detailed examination of daily engagement reveals critical nuances:

- Peak Engagement: Tuesday (123 active days) and Wednesday (106 active days) represent the highest engagement periods
- Mid-Week Consistency: Monday (106) and Thursday (90) maintain strong engagement levels
- Friday Decline: A sharp drop to 73 active days indicates significant end-of-week disengagement
- Weekend Lows: Saturday (78) and Sunday (100) show the lowest engagement, though Sunday demonstrates a slight recovery

Behavioral Implications

The engagement pattern suggests that users perceive TuneWorks as primarily a business workflow tool rather than a platform for weekend activity. The mid-week concentration aligns with core business operations, while the Friday decline reflects the transition out of the work week. The slight Sunday recovery may indicate preliminary week-ahead planning activities.



Recommendations:

Based on the analysis, we recommend immediately aligning product and support strategies with the pronounced weekday engagement pattern. This means concentrating major feature releases, critical communications, and peak customer support staffing from Tuesday through Thursday to capitalize on when users are most active. To counter the significant Friday drop-off, implement "Week in Review" summaries and "Week-Ahead Planning" prompts to help users transition their work and maintain engagement.

For medium-term growth, we advise a targeted initiative to explore and stimulate weekend usage. This involves conducting user research to understand the root causes of low weekend engagement and developing light-touch features, such as mobile-friendly check-ins or Sunday evening planning previews. This strategic approach leverages our core weekday strength while systematically investigating a potential opportunity to expand the product's use cases and overall value.

Note: This analysis reflects total active-day workload, not the number of distinct engagements.

SQL Query:

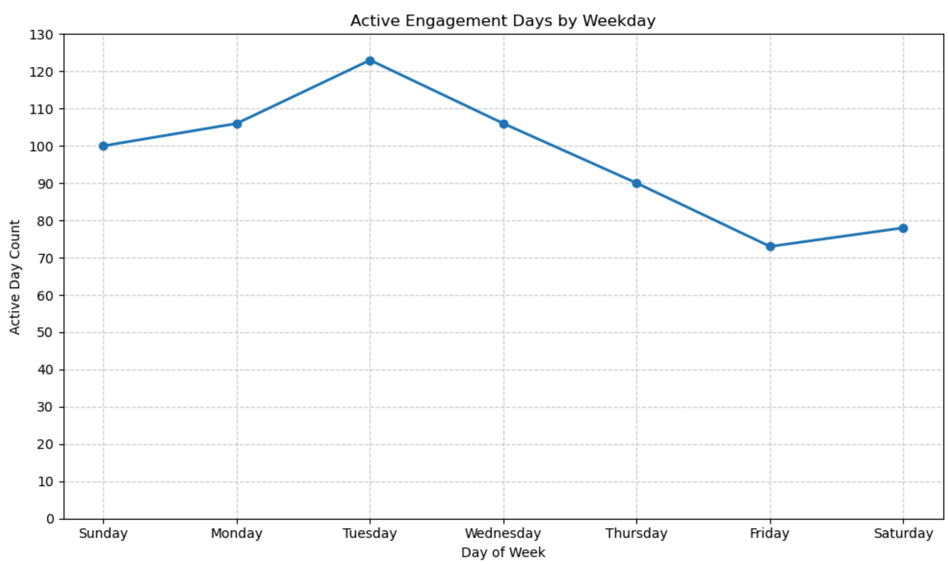
```sql
WITH expanded AS (
  SELECT
     e.engagementnumber,
     d::date AS active_day
  FROM engagements e,
      generate_series(e.startdate, e.enddate, '1 day') AS d
),
raw_counts AS (
  SELECT
     CASE
        WHEN EXTRACT(DOW FROM active_day) IN (0, 6) THEN 'Weekend'
        ELSE 'Weekday'
     END AS day_type,
     COUNT(*) AS active_day_count
  FROM expanded
  GROUP BY day_type
)
SELECT
  day_type,
  active_day_count,
  CASE
     WHEN day_type = 'Weekday' THEN active_day_count / 5.0
     ELSE active_day_count / 2.0
  END AS avg_per_day
FROM raw_counts
ORDER BY day_type DESC;
```

day_type	active_day_count	avg_per_day
Weekend	178	89
Weekday	498	99.6

SQL Query :

```
WITH expanded AS (
  SELECT
     e.engagementnumber,
     d::date AS active_day
  FROM engagements e,
      generate_series(e.startdate, e.enddate, '1 day') AS d
)
SELECT
  TRIM(TO_CHAR(active_day, 'Day')) AS day_name,
  COUNT(*) AS active_day_count
FROM expanded
GROUP BY day_name
ORDER BY MIN(EXTRACT(DOW FROM active_day));
```

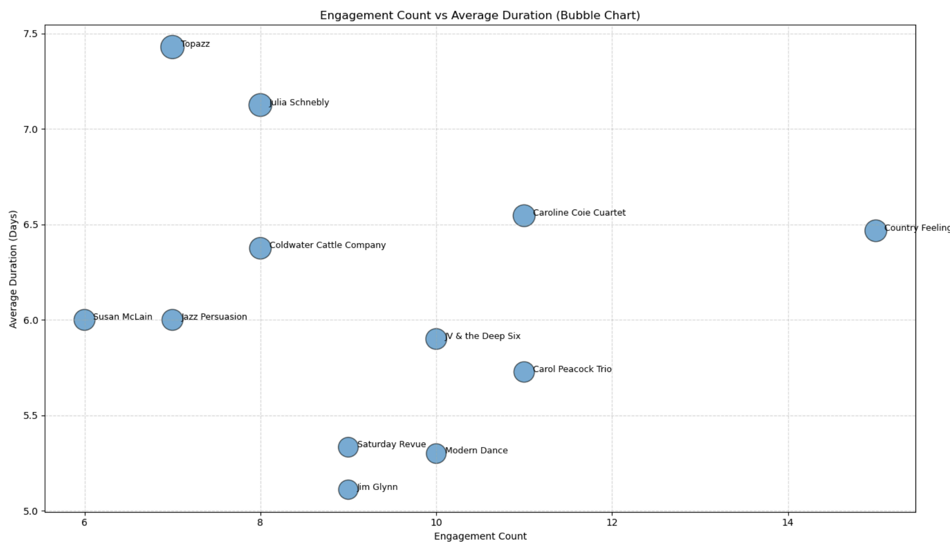
	day_name	active_day_count
1	Sunday	100
2	Monday	106
3	Tuesday	123
4	Wednesday	106
5	Thursday	90
6	Friday	73
7	Saturday	78



4. Do longer engagements tend to involve certain entertainers?

Longer engagements tend to cluster around a subset of entertainers rather than being evenly distributed across all performers. Several entertainers show distinctly higher average engagement durations, notably Topazz, with an average of 7.43 days, and Julia Schnebly, averaging 7.13 days per engagement. Other groups—such as Caroline Coie Cuartet (6.55 days), Country Feeling (6.47 days), and Coldwater Cattle Company (6.38 days)—also appear frequently in multi-day bookings.

In contrast, many entertainers exhibit noticeably shorter average durations, suggesting that extended, week-long engagements may be driven by specific performer characteristics or event formats rather than reflecting general booking behavior. These differences indicate that certain entertainers may specialize in multi-day or residency-style engagements, while others more commonly perform at short-format events.



Recommendations:

Given these patterns, TuneWorks should recognize and strategically leverage entertainers who frequently secure longer, multi-day engagements, as these bookings may offer higher stability and revenue per contract. Marketing and scheduling efforts can be tailored to position long-duration entertainers for extended or residency-style events, while shorter-duration performers can be targeted toward single-day or high-turnover opportunities. Additionally, identifying what drives longer bookings—such as genre, performance style, or client type—may help TuneWorks expand this high-value segment by promoting similar entertainers or designing packages that encourage multi-day engagements.

SQL Query:

```sql
SELECT
  e.entertainerid,
  en.entstagename ,
  AVG(enddate - startdate + 1) AS avg_duration,
  COUNT(*) AS engagement_count
FROM Engagements e
JOIN Entertainers en ON e.entertainerid = en.entertainerid
GROUP BY e.entertainerid, en.entstagename
ORDER BY avg_duration DESC;
```

entertainerid	entstagename	avg_duration	engagement_count
1,002	Topazz	7.4285714286	7
1,011	Julia Schnebly	7.125	8
1,013	Caroline Coie Cuartet	6.5454545455	11
1,008	Country Feeling	6.4666666667	15
1,007	Coldwater Cattle Company	6.375	8
1,012	Susan McLain	6	6
1,005	Jazz Persuasion	6	7
1,003	JV & the Deep Six	5.9	10
1,001	Carol Peacock Trio	5.7272727273	11
1,010	Saturday Revue	5.3333333333	9
1,006	Modern Dance	5.3	10
1,004	Jim Glynn	5.1111111111	9

5. Are there style gaps or underrepresented genres that represent growth opportunities?

If TuneWorks lacks entertainers who specialize in a top customer's preferred genre (e.g., Chamber Music, 60's Music), this points to:

- Recruitment needs
- Portfolio diversification
- Opportunities for targeted talent acquisition

By comparing customer preference counts with the number of entertainers available for each musical style, we observe clear demand–supply gaps. Several genres show higher customer interest than entertainer availability, indicating meaningful growth opportunities.

- Largest gaps (customer demand > entertainer supply):
  40's Ballroom Music (+2), Modern Rock (+2) — customers express interest, but no entertainers can perform these styles.

- Moderate gaps:
  Jazz (+1), Standards (+1), Rhythm & Blues (+1), Classic Rock & Roll (+1), Contemporary (+1) — entertainer supply exists but is insufficient relative to customer demand.

These styles represent areas where TuneWorks could expand talent coverage to better support customer preferences.

Balanced styles:

Genres such as *Salsa, 70's Music, Country Rock, Motown, Show Tunes, Top 40 Hits* show no gap, suggesting that current entertainer coverage adequately meets customer interest.

Oversupplied styles:

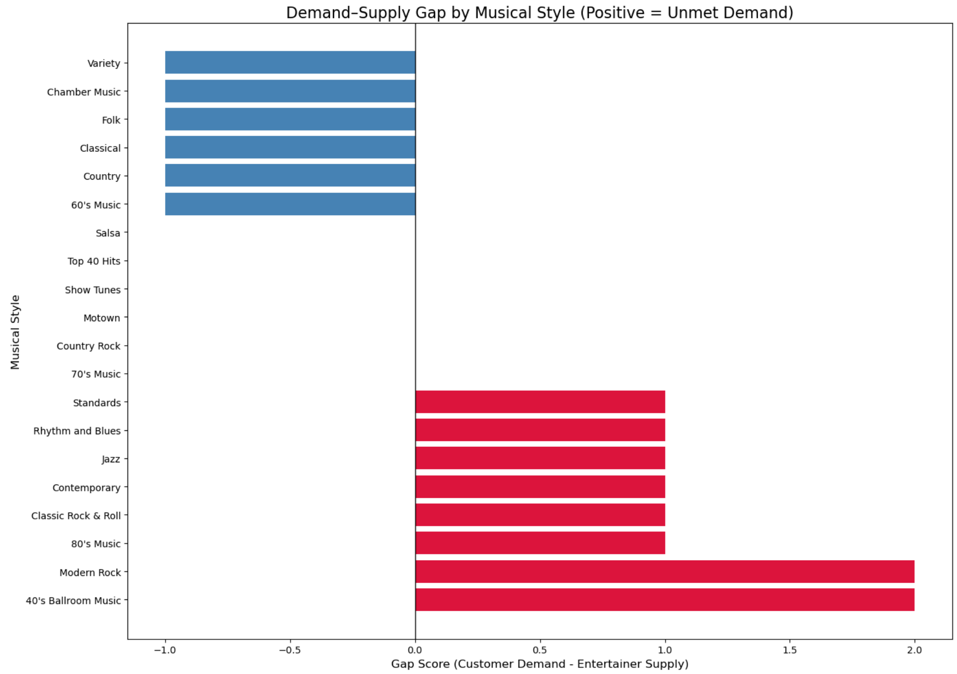
Some genres—*Classical, Variety, Country, Chamber Music, 60's Music, Folk*—have more entertainers than customer demand (gap = −1). These may not require immediate recruitment.

Interpretation:

- Styles with positive gaps represent unmet customer needs and clear recruitment opportunities.
- Filling these gaps can diversify TuneWorks' portfolio and strengthen market competitiveness.
- Oversupplied styles may indicate areas where current resources exceed customer interest.

Limitation:

- Customer preference count measures interest but does not guarantee bookings.
- Entertainer supply reflects capability, not availability or performance quality.



Recommendations:

TuneWorks should prioritize recruiting entertainers in genres where customer interest exceeds current supply—especially **40's Ballroom Music** and **Modern Rock**, where no entertainers are available despite clear demand. Moderate gaps in styles such as **Jazz, Standards, Rhythm & Blues, Classic Rock & Roll, and Contemporary** also represent strong opportunities for targeted talent acquisition and portfolio expansion. Addressing these gaps would better align TuneWorks' offerings with customer preferences and improve competitiveness. Meanwhile, genres with balanced or oversupplied coverage do not require immediate recruitment and can be maintained as-is.

SQL Query:

```
WITH customer_pref AS (
  SELECT mp.styleid, COUNT(*) AS customer_pref_count
  FROM musical_preferences mp
  GROUP BY mp.styleid
),
supply AS (
  SELECT es.styleid, COUNT(DISTINCT es.entertainerid) AS entertainer_supply
  FROM entertainer_styles es
  GROUP BY es.styleid
)
SELECT
```

```sql
  ms.stylename,
  cp.customer_pref_count,
  COALESCE(s.entertainer_supply, 0) AS entertainer_supply,
  (cp.customer_pref_count - COALESCE(s.entertainer_supply, 0)) AS gap_score
FROM customer_pref cp
JOIN musical_styles ms ON cp.styleid = ms.styleid
LEFT JOIN supply s ON cp.styleid = s.styleid
ORDER BY gap_score DESC;
```

stylename	customer_pref_count	entertainer_supply	gap_score
40's Ballroom Music	2	0	2
Modern Rock	2	0	2
Jazz	3	2	1
80's Music	1	0	1
Standards	4	3	1
Rhythm and Blues	3	2	1
Classic Rock & Roll	2	1	1
Contemporary	3	2	1
Salsa	2	2	0
70's Music	1	1	0
Country Rock	1	1	0
Motown	1	1	0
Show Tunes	2	2	0
Top 40 Hits	2	2	0
Classical	2	3	-1
Variety	1	2	-1
Country	1	2	-1
Chamber Music	1	2	-1
60's Music	1	2	-1
Folk	1	2	-1

Customer Analysis

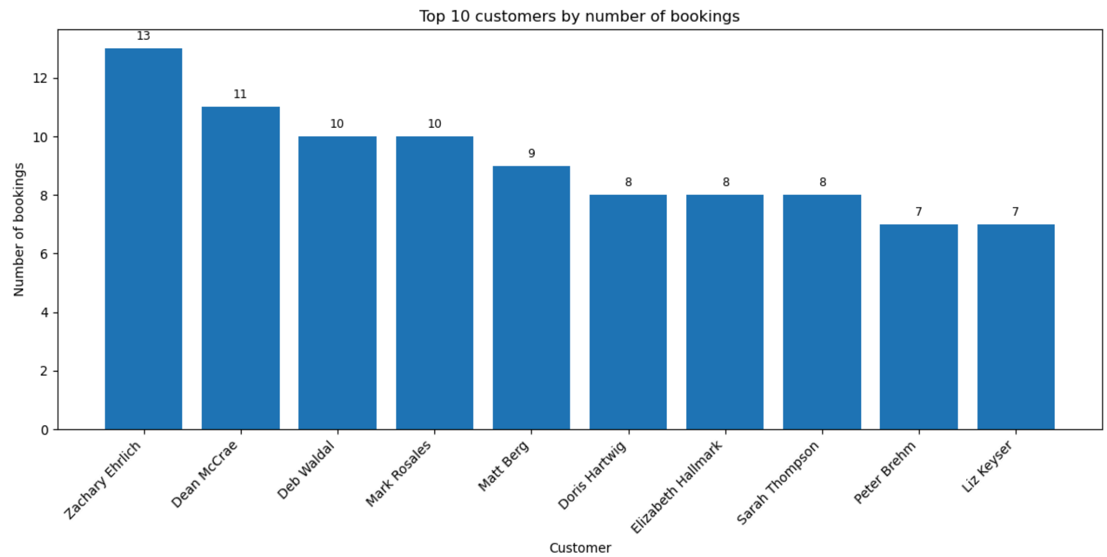
1. Which customers have made the most bookings?

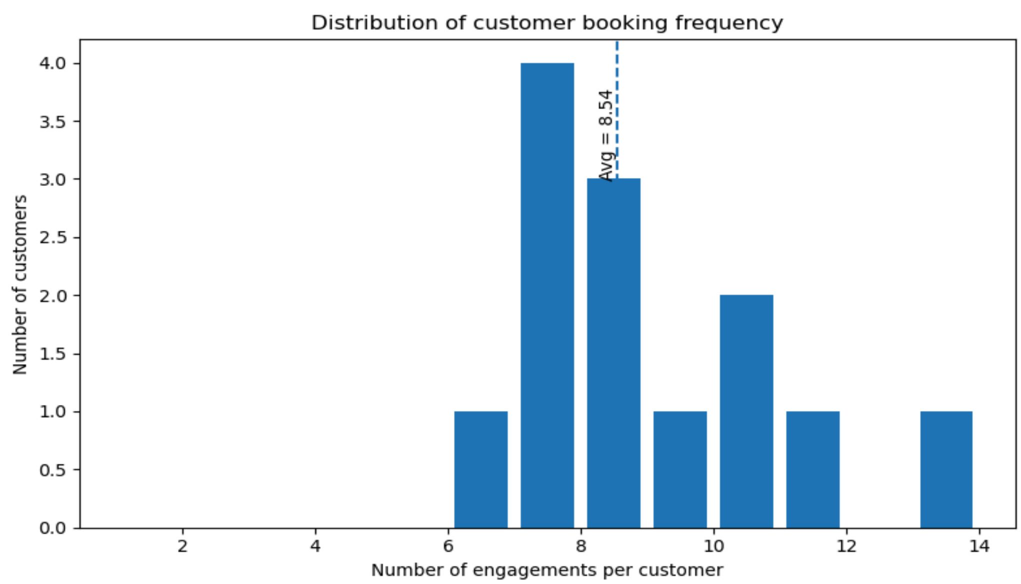
Analysis of customer booking activity shows clear variation in engagement frequency across TuneWorks' client base. A small group of customers accounts for a disproportionately large share of total bookings.

Top customers such as Zachary Ehrlich (13 bookings), Dean McCrae (11), Mark Rosales (10), and Deb Waldal (10) demonstrate high repeat engagement behavior, indicating strong loyalty and potentially ongoing entertainment needs.

A broader set of mid-tier customers—including Matt Berg (9), and Sarah Thompson (8), and two others—also contributes significantly to total booking volume, suggesting multiple repeat interactions rather than one-off engagements.

The long tail of customers shows decreasing booking frequency, with many individuals having only a small number of engagements and some having none at all. This pattern indicates a typical client distribution where a minority of high-value customers drives a large portion of business activity.From a business perspective, this highlights opportunities for **customer segmentation and retention strategy** development.





Recommendations:

TuneWorks should leverage this customer distribution by focusing on differentiated strategies across segments. High-frequency customers—such as those with 10+ bookings—can be targeted with premium offerings, loyalty incentives, or dedicated account management to strengthen retention and maximize lifetime value. Mid-tier customers may benefit from targeted engagement strategies that encourage repeat business and move them into higher-value segments. For low-frequency or inactive customers, re-engagement campaigns, personalized outreach, or promotional packages could help increase activity. Tailoring strategies to each segment will allow TuneWorks to maximize retention, grow customer value, and capture additional bookings across its client base.

SQL Query:

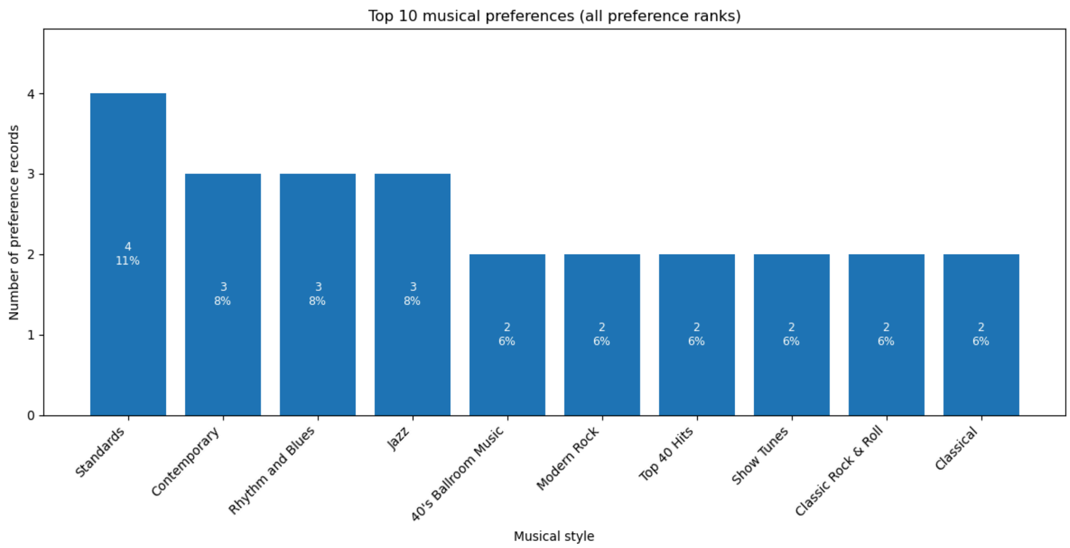
```
SELECT
  c.customerid,
  c.custfirstname,
  c.custlastname,
  COUNT(e.engagementnumber) AS total_bookings
FROM Engagements e
RIGHT JOIN Customers c
  ON e.customerid = c.customerid
GROUP BY
  c.customerid, c.custfirstname, c.custlastname
ORDER BY total_bookings DESC;
```

	customerid	custfirstname	custlastname	total_bookings
1	10,010	Zachary	Ehrlich	13
2	10,004	Dean	McCrae	11
3	10,014	Mark	Rosales	10
4	10,002	Deb	Waldal	10
5	10,006	Matt	Berg	9
6	10,009	Sarah	Thompson	8
7	10,001	Doris	Hartwig	8
8	10,005	Elizabeth	Hallmark	8
9	10,007	Liz	Keyser	7
10	10,003	Peter	Brehm	7
11	10,012	Kerry	Patterson	7
12	10,015	Carol	Viescas	7
13	10,013	Estella	Pundt	6
14	10,008	Darren	Gehring	0
15	10,011	Joyce	Bonnicksen	0

2. What are the most common musical preferences?

Analysis of customer musical preferences shows a diverse range of interests, with several styles emerging as particularly influential in shaping entertainment demand. Customer preferences concentrate around a few key styles—**Standards, Rhythm & Blues, Contemporary, and Jazz**—indicating clear demand clusters that TuneWorks can operationalize.

Overall, the distribution of preferences implies that TuneWorks should maintain a diversified entertainer portfolio, ensuring strong representation across both classic and modern musical styles. The presence of multiple mid-tier genres also indicates opportunities for targeted marketing.



Recommendations:

TuneWorks should leverage these preference patterns by strengthening coverage in the most in-demand styles and proactively matching customers with entertainers who align closely with their preferred genres. Developing genre-specific packages—such as Jazz Night, Contemporary Acoustic Sessions, or Classic Rock Showcases—could help increase booking appeal and better serve customer interests. Ensuring balanced representation across both high-demand and mid-tier genres will enable TuneWorks to capture a broader range of customer needs and enhance overall market responsiveness.

SQL Query:

```
SELECT
  ms.stylename,
  COUNT(*) AS top_preference_count
FROM Musical_Preferences mp
JOIN musical_styles ms
  ON mp.styleid = ms.styleid
GROUP BY ms.stylename
ORDER BY top_preference_count desc
lIMIT 10;
```

	stylename	top_preference_count
1	Standards	4
2	Rhythm and Blues	3
3	Contemporary	3
4	Jazz	3
5	Classic Rock & Roll	2
6	Classical	2
7	Salsa	2
8	40's Ballroom Music	2
9	Modern Rock	2
10	Top 40 Hits	2

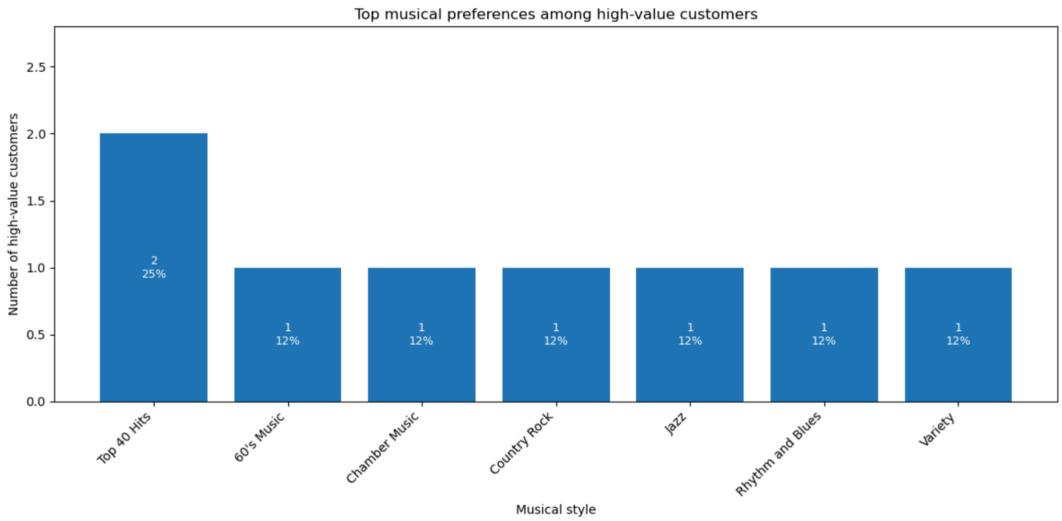
3. What are high-value customers' musical preferences?

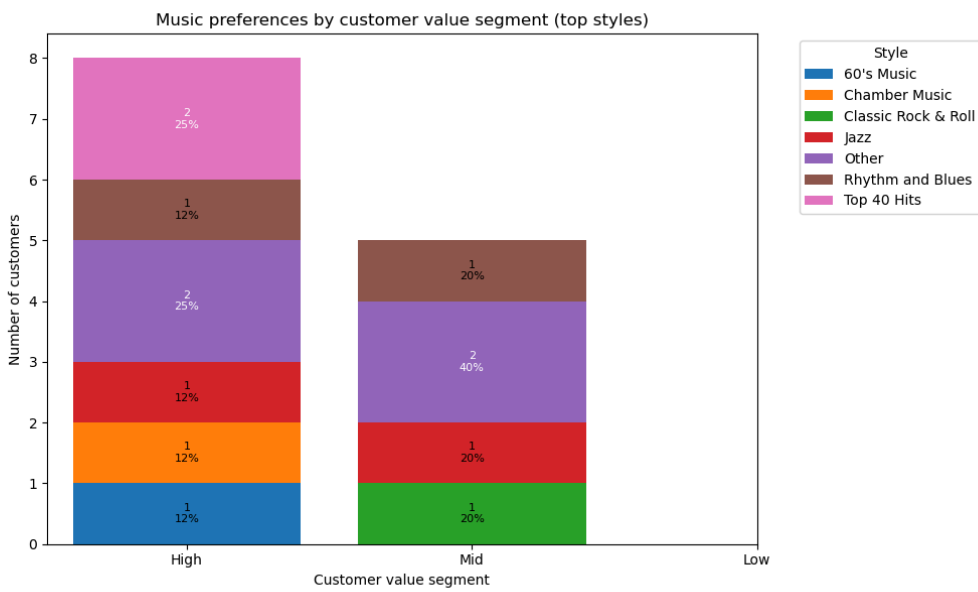
Analysis of the highest-booking customers shows that **top clients(with more than 8 total_bookings) have clear musical preferences, but these preferences span a wide range of genres**—including Rhythm & Blues, Jazz, 60's Music, Top 40 Hits, Variety, Country Rock, and Chamber Music. This diversity suggests that **TuneWorks' most valuable customers do not cluster around a single musical style**, and instead require a broad spectrum of entertainment options.

Furthermore, each high-frequency customer demonstrates a **distinct top preference**, indicating strong individual tastes and offering a natural foundation for personalized engagement strategies.

Overall, the preference patterns reveal three key insights:

- **High-value customers require wide style coverage**—no single genre dominates, so limiting the entertainer portfolio would risk losing top clients.
- **Individualized taste profiles create clear opportunities for personalized matching**, increasing the likelihood of repeated bookings.
- **Certain niche genres (e.g., 60's Music, Chamber Music)** appear among high-frequency clients, signaling potential revenue concentration in less mainstream categories.





SQL Query:

```
WITH high_freq_customers AS (
  SELECT
     customerid,
     COUNT(*) AS total_bookings
  FROM Engagements
  GROUP BY customerid
  HAVING COUNT(*) >= 8
),
customer_top_preferences AS (
  SELECT
     mp.customerid,
     mp.styleid,
```

```
    mp.preferenceseq
  FROM Musical_Preferences mp
  WHERE mp.preferenceseq = 1
)
SELECT
  hfc.customerid,
  c.custfirstname,
  c.custlastname,
  ms.stylename AS top_preference,
  hfc.total_bookings
FROM high_freq_customers hfc
JOIN customer_top_preferences ctp
  ON hfc.customerid = ctp.customerid
JOIN musical_styles ms
  ON ctp.styleid = ms.styleid
JOIN Customers c
  ON hfc.customerid = c.customerid
ORDER BY hfc.total_bookings DESC;
```

	customerid	custfirstname	custlastname	top_preference	total_bookings
1	10,010	Zachary	Ehrlich	Rhythm and Blues	13
2	10,004	Dean	McCrae	Jazz	11
3	10,002	Deb	Waldal	60's Music	10
4	10,014	Mark	Rosales	Top 40 Hits	10
5	10,006	Matt	Berg	Variety	9
6	10,009	Sarah	Thompson	Country Rock	8
7	10,001	Doris	Hartwig	Top 40 Hits	8
8	10,005	Elizabeth	Hallmark	Chamber Music	8

Across Tables Analysis

1. Is there a correlation between entertainer popularity and style strength?

The dataset does not support analyzing a correlation between entertainer popularity and style strength. Entertainers often have multiple musical styles with different StyleStrength values, but engagements do not record which style was performed for each booking. As a result, booking counts reflect entertainer-level popularity only, not style-level popularity. The output table simply repeats the same booking count across all styles an entertainer has, which cannot be interpreted as evidence of any correlation. Because style information is not linked to individual engagements, no meaningful relationship can be drawn between style strength and entertainer popularity.

SQL Query:

```
SELECT p.entertainerid,p.total_bookings,ms.stylename,es.stylestrength
FROM (SELECT entertainerid, COUNT(*) AS total_bookings
  FROM engagements
  GROUP BY entertainerid) p
JOIN entertainer_styles es
ON p.entertainerid = es.entertainerid
join musical_styles ms
on ms.styleid = es.styleid
ORDER BY total_bookings DESC;
```

entstagename	total_bookings	stylename	stylestrength
Country Feeling	15	Country	1
Country Feeling	15	60's Music	2
Caroline Coie Cuartet	11	Jazz	1
Caroline Coie Cuartet	11	Contemporary	2
Carol Peacock Trio	11	Contemporary	2
Carol Peacock Trio	11	Show Tunes	1
Carol Peacock Trio	11	Standards	3
Modern Dance	10	Variety	1
JV & the Deep Six	10	Classic Rock & Roll	2
JV & the Deep Six	10	60's Music	1
Modern Dance	10	Top 40 Hits	3
Modern Dance	10	Salsa	2
Saturday Revue	9	70's Music	2
Saturday Revue	9	Standards	3
Saturday Revue	9	Top 40 Hits	1
Jim Glynn	9	Folk	1
Julia Schnebly	8	Classical	2
Coldwater Cattle Company	8	Country Rock	1
Julia Schnebly	8	Chamber Music	1

Julia Schnebly	8	Show Tunes	3
Coldwater Cattle Company	8	Country	2
Jazz Persuasion	7	Rhythm and Blues	1
Jazz Persuasion	7	Salsa	2
Topazz	7	Motown	2
Jazz Persuasion	7	Jazz	3
Topazz	7	Variety	3
Topazz	7	Rhythm and Blues	1
Susan McLain	6	Folk	1
Susan McLain	6	Classical	2

2. Are certain agents more likely to book specific entertainers or musical styles?

Overall, the data shows that some agents do appear more likely to book certain entertainers, but the drivers behind these patterns differ. Some high booking counts occur simply because those entertainers are generally very popular across all agents, while a few cases reflect genuine agent-specific preferences. Style-level preferences, however, cannot be reliably analyzed due to data limitations.

At the entertainer level, when filtering for pairs with more than two bookings and comparing them against overall entertainer popularity, we observe that many high-frequency agent–entertainer combinations involve entertainers who are already top-ranked in total bookings (such as Entertainers 1008, 1001, and 1013). These cases reflect the entertainer's global popularity rather than a true agent preference. However, a small number of combinations—such as Agent 3 with Entertainer 1007 and Agent 8 with Entertainer 1004—show disproportionately high booking counts for entertainers who are not top overall performers, indicating real agent-specific preferences.

At the musical style level, no meaningful conclusions can be drawn. Engagements record only entertainer IDs and do not specify which musical style was performed. Since entertainers may have multiple styles, and no engagement-level style information exists, style-based preference analysis is not possible.

In summary, certain agents do exhibit clear booking preferences for specific entertainers, but these patterns must be interpreted alongside overall entertainer popularity. Musical style preferences cannot be evaluated with the current dataset.

SQL Query:

```
WITH overall_popularity AS (
  SELECT
    entertainerid,
    COUNT(*) AS total_bookings
  FROM engagements
  GROUP BY entertainerid
),
agent_preferences AS (
  SELECT
    a.agentid,
    a.agtfirstname || ' ' || a.agtlastname AS agent_name,
    e.entertainerid,
    en.entstagename,
    COUNT(*) AS agent_booking_count
  FROM engagements e
  JOIN agents a ON e.agentid = a.agentid
  JOIN entertainers en ON e.entertainerid = en.entertainerid
  GROUP BY a.agentid, agent_name, e.entertainerid, en.entstagename
  HAVING COUNT(*) > 2
)
SELECT
  ap.agent_name,
  ap.entertainerid,
  ap.entstagename,
  ap.agent_booking_count,
  op.total_bookings AS entertainer_overall_count
FROM agent_preferences ap
JOIN overall_popularity op
  ON ap.entertainerid = op.entertainerid
ORDER BY ap.agentid, ap.agent_booking_count DESC;
```

agent_name	entertainerid	entstagename	agent_booking_count	entertainer_overall_count
William Thompson	1,003	JV & the Deep Six	3	10
Carol Viescas	1,006	Modern Dance	3	10
Carol Viescas	1,008	Country Feeling	3	15
Karen Smith	1,011	Julia Schnebly	3	8
Karen Smith	1,008	Country Feeling	3	15
Marianne Wier	1,001	Carol Peacock Trio	4	11
Marianne Wier	1,013	Caroline Coie Cuartet	3	11
John Kennedy	1,008	Country Feeling	3	15
Caleb Viescas	1,008	Country Feeling	3	15
Maria Patterson	1,001	Carol Peacock Trio	4	11
Maria Patterson	1,010	Saturday Revue	3	9
Maria Patterson	1,004	Jim Glynn	3	9

SQL Query:

```sql
SELECT entertainerid, COUNT(*)
FROM Engagements
GROUP BY entertainerid
ORDER BY count DESC;
```

entertainerid	count
1,008	15
1,013	11
1,001	11
1,006	10
1,003	10
1,010	9
1,004	9
1,011	8
1,007	8
1,002	7
1,005	7
1,012	6